ASSIGNMENT No. 06

Multidimensional Arrays: *Implement matrix operations including addition, subtraction, multiplication, transpose, diagonal sum, symmetric matrix check, and sparse matrix representation.*

1. Nirav Modi is your friend's Uncle who works as a Bank Manager in the US. He has visited your home for the summer holidays with his family. One day, he invites you and your friend to solve a banking problem. He has a list of X number of customers with both positive and negative balances. He tells you that he has decided to quit his bank job and return to India, as the CEO of the Bank that has made a real bad joke about him. Before he leaves the job, he wants to alter the customer balances in the following way to get revenge on the Bank.

   For the first customer, he wants to replace the balance as the absolute difference between the total bank balances of people with positive balances and the total balance of people with negative balances. For 1st customer, this is computed from 2nd to the last customer. Similarly, for the 2nd customer, he wants to replace the balance as the absolute difference between the sum of positive balance and negative balance between the 3rd and last customer. For the last customer in the list, the replacement will be 0.

   For example, if the list contains X = 5 customers where balance is = [1, -1, 2, 3, -2], the new balance will be [2, 3, 1, 2, 0]. If the list contains X = 6 customers where balance is = [-3, -4, -2, 5, 1, -2], where the new balance will be [2, 2, 4, 1, 2, 0]. Uncle Nirav has promised your friend and you that if you two guys solve this issue, he will buy you a Piggy Bank for your savings.

2. Lord Voldemort, a former employee at Walmart, is a famous thief who recently got fired for stealing packs of KitKats from the candy section of his store. After being released from jail, he wants to pursue a full-time career in stealing chocolates, not from bigger stores but from smaller homes. So, he picks up a street from Dabra Village, containing N rows and M columns of houses. He wants to start the stealing process from house 1 in the upper left of the street and travel in the straight of the row. When he finds that the row of houses is coming to an end, he only takes the right. Then again, start visiting all the homes until he reaches the lower right of the street. He further takes only right turns. Finally, he visits the last home in the centre of the street to finish his job. Now as an experienced detective, you need to track the order of houses he stole in the street.

   If the street contains 4 rows and 5 columns then houses are in the order:
   1 2 3 4
   5 6 7 8
   9 10 11 12
   13 14 15 16
   Output: 1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10.
   If the street contains 3 rows and 6 columns then houses are in the order:
   1 2 3 4 5 6
   7 8 9 10 11 12

13 14 15 16 17 18
Output: 1 2 3 4 5 6 12 18 17 16 15 14 13 7 8 9 10 11

3. Given a square matrix of size n × n (n ≤ 10), write a C-code to rotate the matrix 90 degrees clockwise without using any auxiliary matrix.
   Hint: See the example below Input:

$$A = \begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{matrix}$$

   Output (after 90° clockwise rotation):

$$A' = \begin{matrix} 7 & 4 & 1 \\ 8 & 5 & 2 \\ 9 & 6 & 3 \end{matrix}$$

4. Write a C program that extracts and displays the boundary elements of a matrix. The boundary elements are those that lie along the outer edges of the matrix, specifically the elements in the first and last rows, as well as the elements in the first and last columns. The program should read the matrix from the user, identify these boundary elements, and display them in a clear format.

5. Write a program that checks whether a given square matrix is a diagonal matrix. A diagonal matrix is defined as a square matrix in which all the non-diagonal elements are zero, and only the elements on the main diagonal (from the top-left to the bottom-right) may be non-zero. The program should read the matrix from the user, examine its elements, and determine whether it satisfies the conditions of a diagonal matrix.

6. Write a C program that checks whether two matrices A and B commute. Two matrices are said to be commutative if the product of A and B is equal to the product of B and A, i.e., A × B = B × A. The program should read both matrices, perform the multiplications, and then compare the resulting matrices to determine whether they commute.